

Day 6 Task – Submitted By 22-06-2026

Logistic Regression (Supervised) vs K-Means Clustering (Unsupervised)

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1. Setup & Imports

Libraries used: pandas for data handling, scikit-learn for ML models, matplotlib and seaborn for visualization, and PCA/silhouette_score for cluster evaluation.

Code:

```
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.cluster import KMeans
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
```

2. Load Dataset

The Titanic dataset is loaded using pandas. It contains 891 passengers with 12 columns including survival status, age, class, fare, and gender.

Code:

```
titanic = pd.read_csv("Titanic-Dataset.csv")
titanic.head(5)
```

Output – First 5 Rows:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

3. Descriptive Statistics

Code:

```
titanic.describe()
```

Output:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

4. Dataset Info

Code:

```
titanic.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
10   Cabin        204 non-null    object
11   Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

5. Missing Values

Code:

```
titanic.isna().sum()
```

Output – Before Imputation:

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64
```

6. Data Preprocessing

Missing Age values (177) are filled with the median age to preserve the dataset size without introducing bias.

Code:

```
titanic['Age'] = titanic['Age'].fillna(titanic['Age'].median())
```

Output – After Imputation:

```
Age    0 (177 missing values filled with median)
```

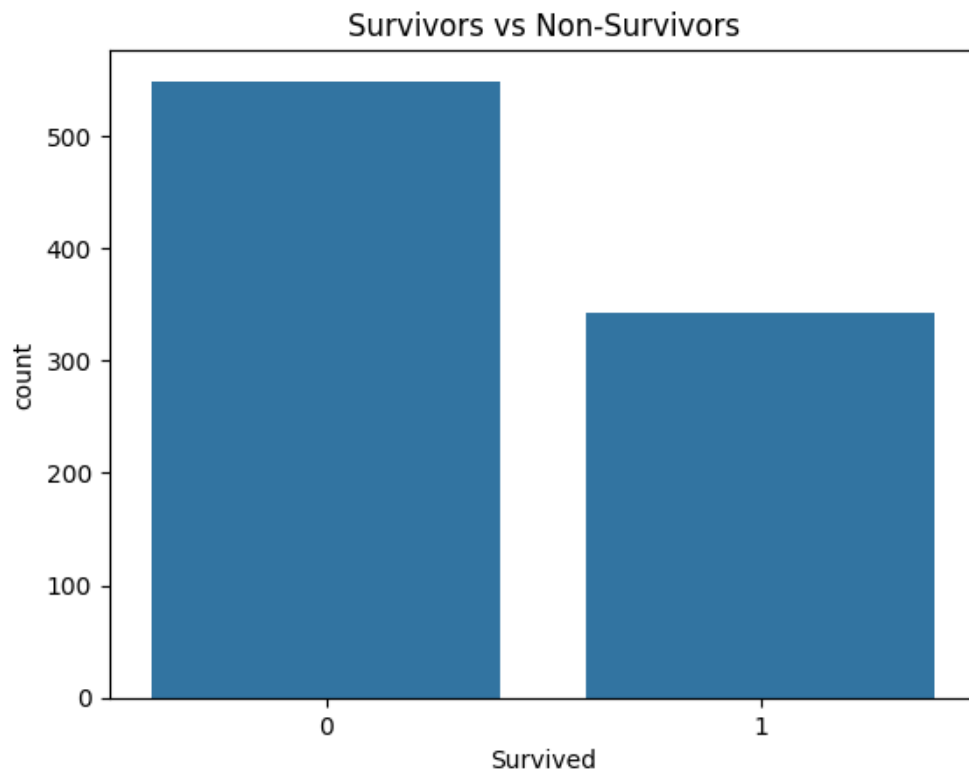
7. Exploratory Data Analysis (EDA)

7.1 Survival Count

Code:

```
sns.countplot(x='Survived', data=titanic)
plt.title('Survivors vs Non-Survivors')
```

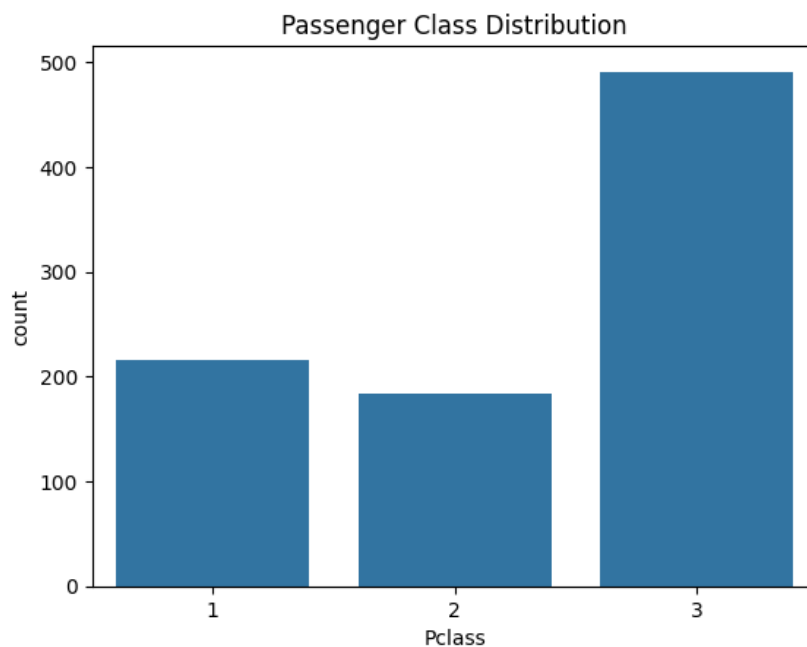
```
plt.show()
```



7.2 Passenger Class Distribution

Code:

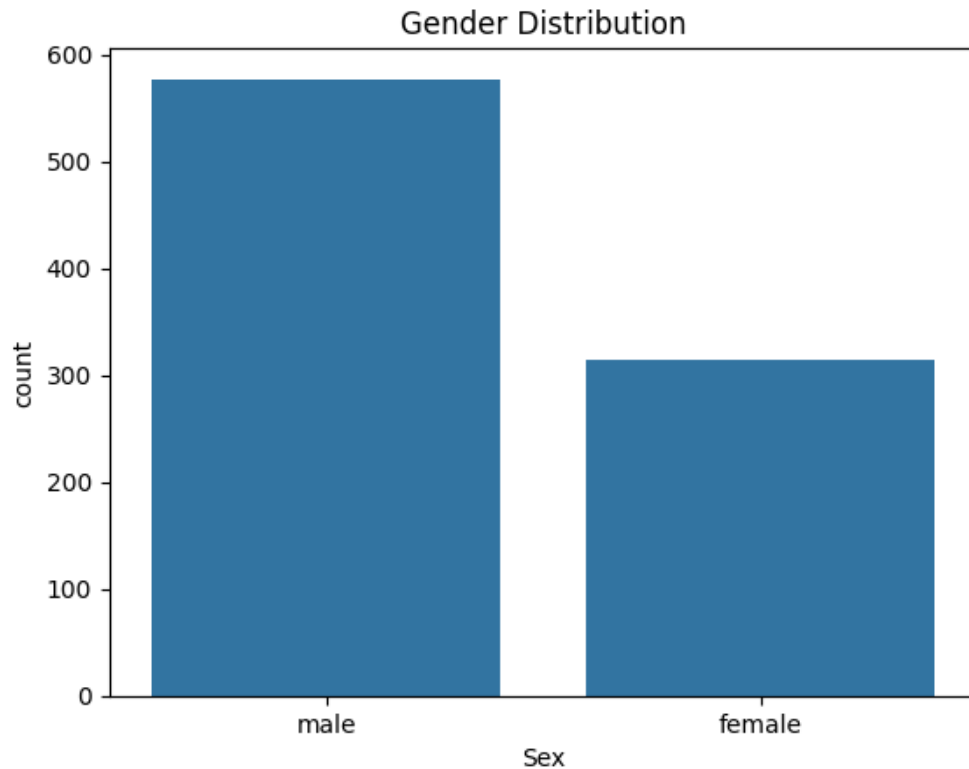
```
sns.countplot(x='Pclass', data=titanic)  
plt.title('Passenger Class Distribution')  
plt.show()
```



7.3 Gender Distribution

Code:

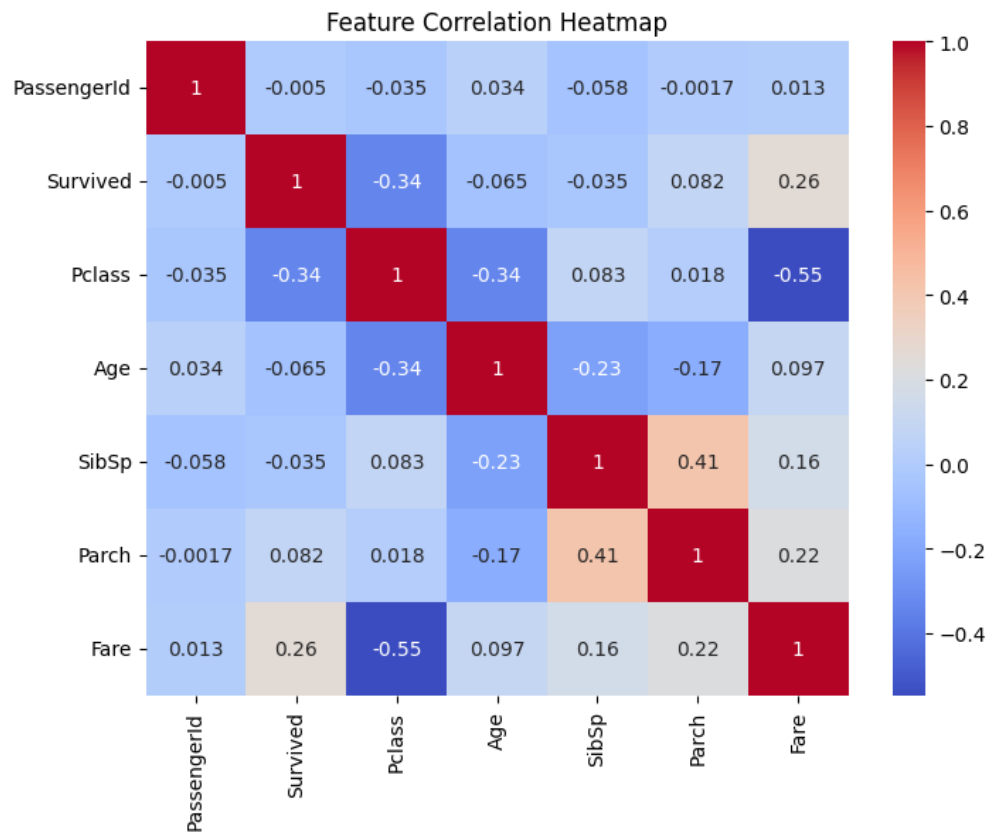
```
sns.countplot(x='Sex', data=titanic)
plt.title('Gender Distribution')
plt.show()
```



7.4 Feature Correlation Heatmap

Code:

```
numeric_cols = titanic.select_dtypes(include=['number'])
sns.heatmap(numeric_cols.corr(), annot=True, cmap='coolwarm')
plt.title('Feature Correlation Heatmap')
plt.show()
```



8. Feature Engineering & Train-Test Split

Code:

```
X = titanic[['Age', 'Pclass', 'Sex']]
y = titanic['Survived']
X['Sex'] = X['Sex'].map({'male': 0, 'female': 1})
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

9. Part 1: Supervised Learning – Logistic Regression

Goal: Predict passenger survival using Logistic Regression. Features used: Age, Pclass, Sex (encoded). 80% training data, 20% test data.

Code:

```
log_model = LogisticRegression(max_iter=200)
log_model.fit(X_train, y_train)
y_pred = log_model.predict(X_test)
```

```
print('Supervised Learning (Logistic Regression):')
print(f'Accuracy: {accuracy_score(y_test, y_pred):.2f}')
print('Classification Report:', classification_report(y_test, y_pred))
```

Output:

```
Supervised Learning (Logistic Regression):
Accuracy: 0.81
Classification Report:
              precision    recall  f1-score   support

     0       0.82        0.87        0.84        105
     1       0.79        0.73        0.76         74

 accuracy          0.81
 macro avg         0.81        0.80        0.80        179
weighted avg         0.81        0.81        0.81        179
```

9.1 Confusion Matrix

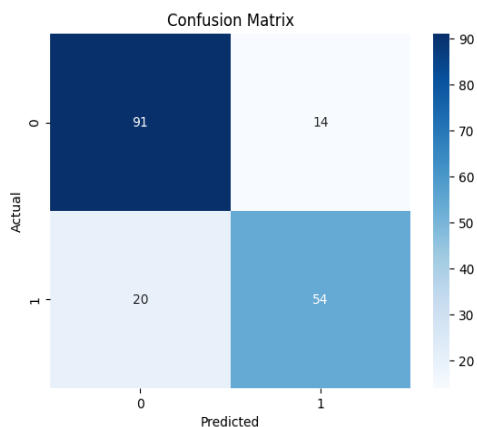
Code:

```
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted') plt.ylabel('Actual')
plt.title('Confusion Matrix') plt.show()
```

Raw Output:

Confusion Matrix:

```
[[91 14]
 [20 54]]
```



10. Part 2: Unsupervised Learning – K-Means Clustering

Goal: Group passengers by Age, Fare, and Pclass without using survival labels. This reveals natural clusters based on passenger characteristics.

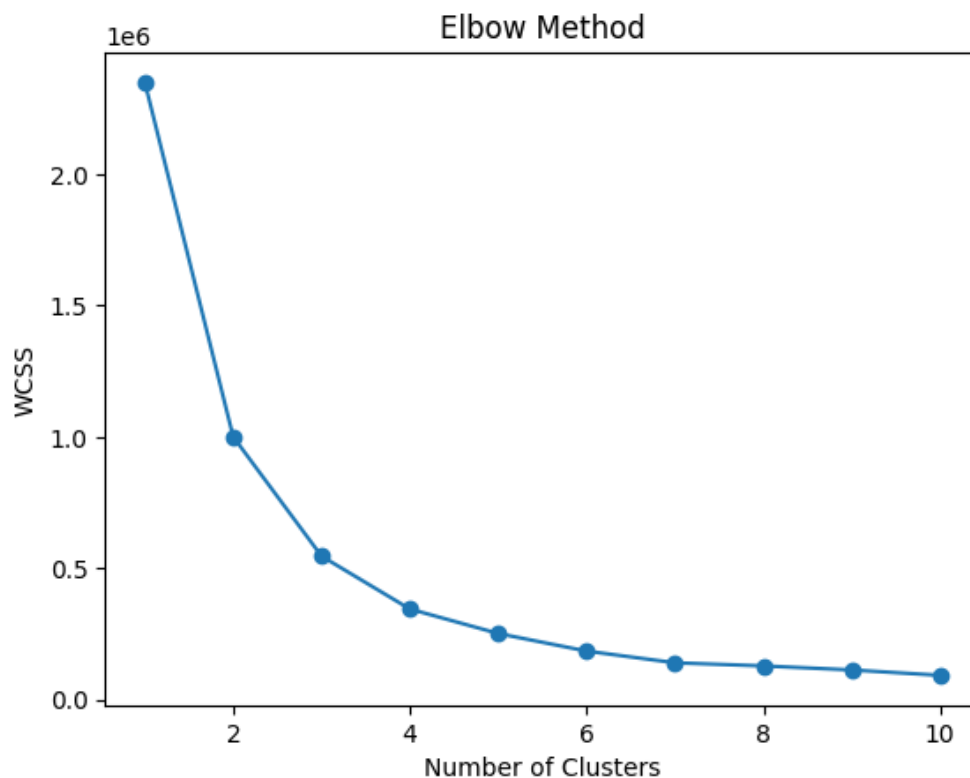
Code:

```
X_cluster = titanic[['Age', 'Fare', 'Pclass']]
kmeans = KMeans(n_clusters=3, random_state=42)
titanic['Cluster'] = kmeans.fit_predict(X_cluster)
```

10.1 Elbow Method – Choosing k=3

Code:

```
wcss = []
for i in range(1, 11):
    km = KMeans(n_clusters=i, random_state=42)
    km.fit(X_cluster)
    wcss.append(km.inertia_)
plt.plot(range(1,11), wcss, marker='o')
plt.title('Elbow Method')
plt.show()
```



10.2 Cluster Centers

Output:

Unsupervised Learning (K-Means clustering)

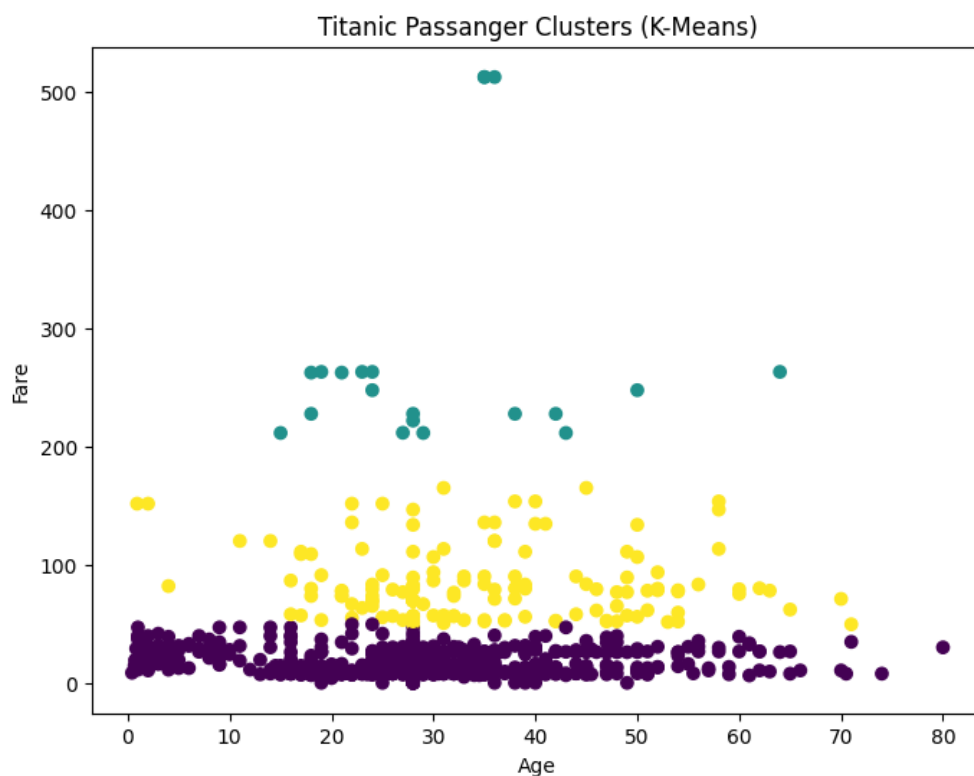
Cluster Centers:

```
[[ 28.28 15.45  2.55] <- Cluster 0: Young, low fare, 3rd class  
[ 30.85 279.31  1.00] <- Cluster 1: Middle-aged, high fare, 1st class  
[ 34.71  83.39  1.25]] <- Cluster 2: Older, mid fare, 1st/2nd class
```

10.3 Cluster Scatter Plot (Age vs Fare)

Code:

```
plt.scatter(titanic['Age'], titanic['Fare'], c=titanic['Cluster'], cmap='viridis')  
plt.xlabel('Age') plt.ylabel('Fare')  
plt.title('Titanic Passenger Clusters (K-Means)')  
plt.show()
```

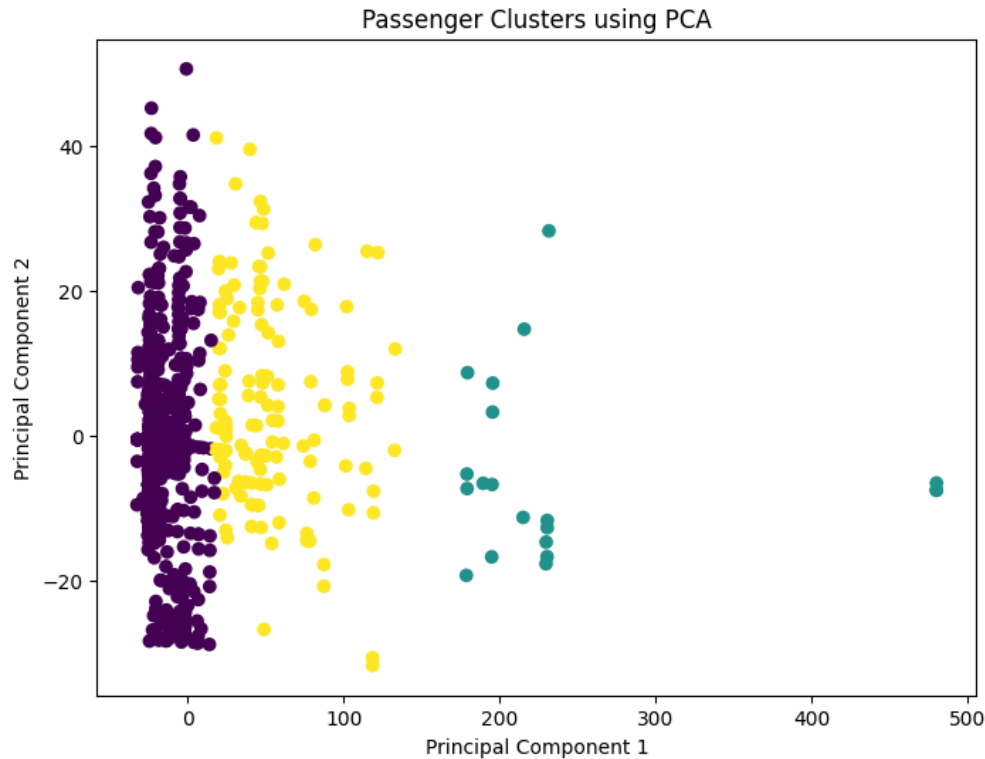


10.4 PCA Visualization

Code:

```
pca = PCA(n_components=2)  
reduced = pca.fit_transform(X_cluster)
```

```
plt.scatter(reduced[:,0], reduced[:,1], c=titanic['Cluster'], cmap='viridis')
plt.title('Passenger Clusters using PCA')
plt.show()
```



10.5 Silhouette Score

Output:

Silhouette Score: 0.664

A silhouette score of 0.664 indicates well-separated, cohesive clusters — a good result for $k=3$.

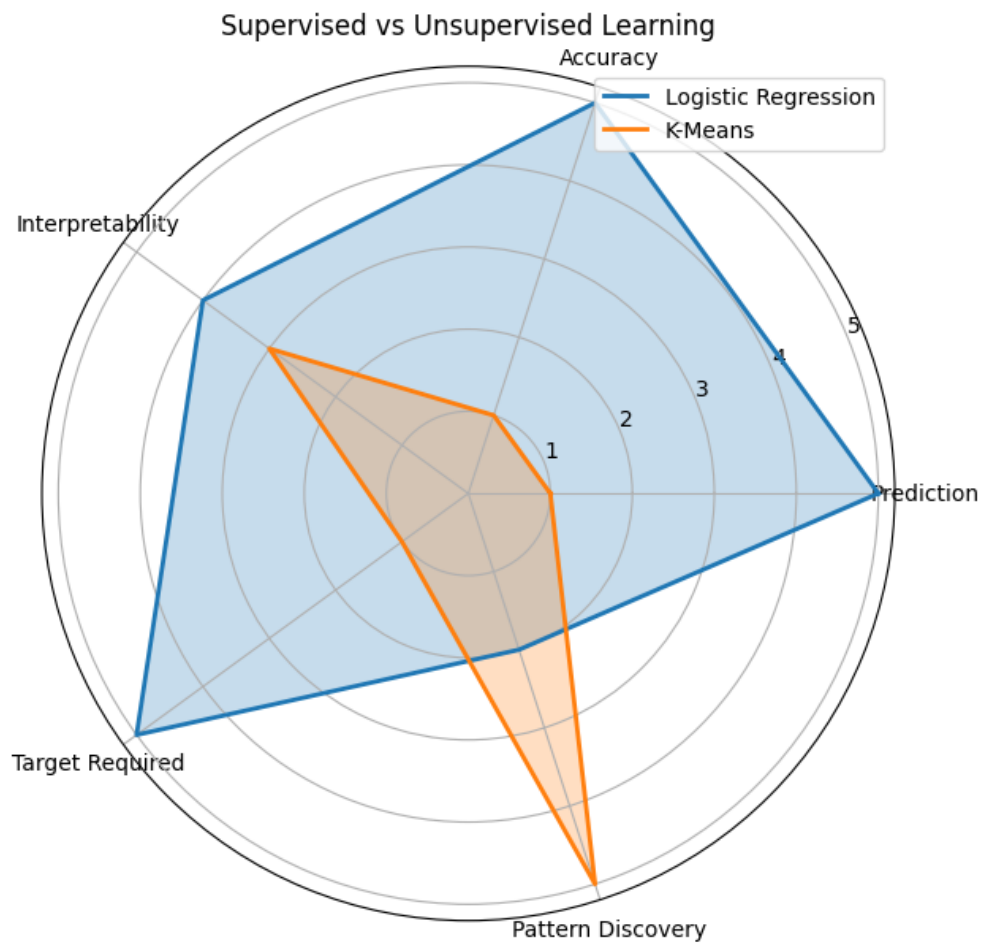
11. Comparison: Supervised vs Unsupervised

A radar chart comparing Logistic Regression and K-Means across five capability dimensions: Prediction, Accuracy, Interpretability, Target Required, and Pattern Discovery.

Code:

```
categories = ['Prediction','Accuracy','Interpretability','Target Required','Pattern
Discovery']
logistic = [5,5,4,5,2]
kmeans = [1,1,3,1,5]
# Polar plot with fill ...
```

```
plt.title('Supervised vs Unsupervised Learning')
plt.show()
```



12. Summary Comparison Table

Criteria	Logistic Regression	K-Means Clustering
Type	Supervised	Unsupervised
Goal	Predict survival (binary)	Group passengers naturally
Features	Age, Pclass, Sex	Age, Fare, Pclass
Accuracy	81%	N/A (no labels)
Silhouette Score	N/A	0.664 (good)
Target Required	Yes (Survived)	No
Output	0 or 1 per passenger	Cluster 0, 1, or 2
Best Use Case	Classification / prediction	Pattern discovery

13. Conclusion

This task demonstrated two fundamental ML paradigms on the Titanic dataset:

Logistic Regression (Supervised): Achieved 81% accuracy in predicting survival. The model requires labeled data and outputs a direct prediction per passenger.

K-Means Clustering (Unsupervised): Grouped passengers into 3 meaningful clusters based on age, fare, and class — without any survival labels. Silhouette score of 0.664 confirms well-separated clusters.

Key takeaway: Use Supervised Learning when you have labeled data and want to predict outcomes; use Unsupervised Learning to discover hidden structure in unlabeled data.